# Samuel Mensah

*Financial Analyst — Credit and Market Risk*

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## PROFESSIONAL SUMMARY

Risk analyst with 13 years of experience in credit risk, market risk, and microfinance portfolio monitoring across South Africa and Ghana. Develops and monitors IFRS 9 expected credit loss models, probability of default, loss given default, and exposure at default models, stress tests, and Value at Risk measures. Currently maintains expected credit loss models for a ZAR 48 billion retail unsecured portfolio and integrates macroeconomic scenarios into stress testing. Builds model monitoring and early warning dashboards in SQL, Python, SAS, and Power BI, with documentation that supports internal model validation and audit requests. Communicates risk findings clearly to credit committees and business teams, and is interested in climate risk and explainable credit models.

## ANALYTICAL AND TECHNICAL SKILLS

**Risk management:** Credit risk, market risk, IFRS 9 expected credit loss (ECL), stress testing, Value at Risk (VaR) and expected shortfall, limit monitoring, early warning indicators, climate risk scenario analysis.

**Quantitative methods:** Logistic regression scorecards, PD, LGD, and EAD modelling, survival analysis, Monte Carlo and historical simulation, model performance metrics (Gini, KS, PSI), backtesting.

**Financial modelling:** Macroeconomic scenario modelling, portfolio loss forecasting, provisioning and capital impact analysis.

**Accounting and financial statements:** IFRS 9 staging and provisioning, credit analysis of SME financial statements.

**Programming and data analytics:** Advanced: SQL, Python (pandas, scikit-learn, statsmodels), SAS. Working knowledge: R.

**Business intelligence and reporting:** Power BI, Excel, credit committee packs, regulatory and management reporting support.

**Model governance:** Model documentation, validation findings management, Basel framework concepts.

**Industry experience:** Retail and commercial banking, investment banking, microfinance.

## PROFESSIONAL EXPERIENCE

**Highveld Commercial Bank** (retail and commercial banking) — Johannesburg, South Africa
*Senior Credit Risk Analyst* — *Mar 2020 – Present*

- Maintain IFRS 9 expected credit loss models for a ZAR 48 billion retail unsecured lending portfolio, including staging rules, PD term structures, and LGD estimates; present monthly provision movements to the credit committee.
- Led the integration of three macroeconomic scenarios into ECL and stress testing models with the economics team, giving management a clearer view of provisions in a severe downturn.
- Built monitoring for 11 credit models with Gini, PSI, and calibration tests in Python and SQL; two models with deteriorating performance were redeveloped as a result.
- Developed a Power BI early warning dashboard combining arrears, credit bureau data, and behavioural signals, which helped collections teams prioritise accounts and reduced 90-day roll rates by about 8% in a pilot segment.
- Prepare model documentation and evidence packs for internal model validation and external audit requests.

**Gold Coast Investment Bank** (investment banking) — Accra, Ghana
*Market Risk Analyst* — *Jan 2016 – Feb 2020*

- Produced daily Value at Risk and expected shortfall for fixed income and FX trading books using historical simulation, and reported limit utilisation and breaches to the head of markets.
- Built VaR backtesting in Python and documented exceptions, which supported a review of the model's lookback window.
- Designed stress scenarios for currency depreciation and interest rate shocks with the treasury team.
- Automated daily risk reports in SQL and Excel, reducing preparation time from three hours to under one hour.



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**Akwaaba Microfinance** (microfinance) Kumasi, Ghana
*Risk Analyst* *Sep 2013 – Dec 2015*

- Produced portfolio-at-risk reports for about 40,000 microloans across 22 branches.
- Built an application scorecard with logistic regression that reduced first-payment default in a pilot branch.
- Trained branch staff on risk reporting and data entry quality.

## SELECTED PROJECTS

**Climate risk scenario analysis for SME lending**
*Tools and methods: Python, SQL, sector and regional scenario mapping, PD overlays*

- **Problem:** The bank had no view of how drought and transition risk could affect agricultural and transport SME borrowers.
- **Contribution:** Mapped exposures by sector and region and built scenario-based PD adjustments with the sustainability team.
- **Result:** Identified ZAR 2.1 billion of exposure in high-sensitivity segments for further review.

**Behavioural scorecard redevelopment**
*Tools and methods: SAS, Python, scikit-learn, SHAP explainability*

- **Problem:** The legacy scorecard had lost discriminatory power, with Gini falling from 0.62 to 0.48.
- **Contribution:** Rebuilt features and the model, with SHAP-based explanations prepared for credit committee review.
- **Result:** Gini returned to 0.61 on out-of-time validation.

**VaR engine for graduate training**
*Tools and methods: Python, NumPy, pandas, historical, parametric, and Monte Carlo VaR*

- **Problem:** Graduate analysts learned VaR from spreadsheets with hidden logic.
- **Contribution:** Wrote a documented library covering three VaR methods with worked examples.
- **Result:** Used in the bank's graduate risk training programme.

## EDUCATION

**Rand Institute of Finance** Johannesburg, South Africa (distance learning)
*M.Sc. Financial Risk Management (part-time)* *2017 – 2019*

- Dissertation: macroeconomic sensitivity of unsecured lending defaults in emerging markets.
- Coursework: credit risk modelling, market risk, derivatives, financial econometrics.

**Ashanti University of Applied Sciences** Kumasi, Ghana
*B.Sc. Actuarial Science* *2009 – 2013*

- Coursework: probability, statistics, financial mathematics, survival models.

## HACKATHONS, CONFERENCES, AND PROFESSIONAL ACTIVITIES

- **African Risk Analytics Conference, Johannesburg (2025)** — Speaker on monitoring IFRS 9 models in volatile economies.
- **Credit Risk Modelling Hackathon, Cape Town (2022)** — Built a default prediction model on an anonymised dataset with a team of four, with a focus on explainability for credit decisions.
- **Risk analytics mentoring (2021 – Present)** — Mentors graduate analysts in the bank's risk development programme.

## CERTIFICATIONS AND ADDITIONAL QUALIFICATIONS

**Professional development (illustrative):** Credit Risk Modelling Programme, Rand Institute of Finance (2021); Climate Risk Fundamentals Course, Northbridge Finance Institute (2023).

**Languages:** English (fluent), Twi (native), Afrikaans (basic).


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